54.2: Diversity and trophic structure characterize biological communities

Ecological communities can be characterized by certain general attributes, including how diverse they are and the feeding relationships of their species. In some cases, as you'll read, a few species exert strong control on a community's *structure*—the number, identity, and relative abundance of its species.

Species Diversity

The **species diversity** of a community—the variety of different kinds of organisms that make up the community—has two components. One is **species richness**, the number of different species in the community. The other is the **relative abundance** of the different species, the proportion each species represents of all individuals in the community.

Imagine two small forest communities, each with 100 individuals distributed among four tree species (A, B, C, and D) as follows:

Community 1: 25A,25B,25C,25DCommunity 2: 80A,5B,5C,10D

The species richness is the same for both communities because they both contain four species of trees, but the relative abundance is very different **(Figure 54.12)**. You would easily notice the four types of trees in community 1, but unless you looked carefully, you might see only the abundant species A in the second forest. Most observers would intuitively describe community 1 as the more diverse of the two communities.

Figure 54.12

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Which forest is more diverse?
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Tree species labeled A, B, C, D. Community 1: 25% species A; 25% species B; 25% species C and 25% species D. Community 2: 80% species A; 5% species B; 5% species C and 10% species D.

Ecologists would say that community 1 has greater species diversity, a measure that includes both species richness and relative abundance.

Ecologists use many tools to compare the diversity of communities. They often calculate indexes of diversity based on species richness and relative abundance. One widely used index is the **Shannon diversity index** (H):

H=-pAlnpA+pBlnpB+pClnpC+...

where A,B,C... are the species in the community, p is the relative abundance of each species, and ln is the natural logarithm; the ln of each value of p can be determined using the “ln” key on a scientific calculator. A higher value of H indicates a more diverse community. Let’s use this equation to calculate the Shannon diversity index of the two communities in Figure 54.12. For community 1, p=0.25 for each species, so

H=−4(0.25 ln 0.25)=1.39.

For community 2,

H=−[0.8 ln 0.8+2(0.05 ln 0.05)+0.1 ln 0.1]=0.71.

These calculations confirm our intuitive description of community 1 as more diverse.

Determining the number and relative abundance of species in a community can be challenging. Because most species in a community are relatively rare, it may be hard to obtain a sample size large enough to be representative. It can also be difficult to identify some of the species in the community. If an unknown organism cannot be identified on the basis of morphology alone, it is useful to compare all or part of its genome to a reference database of DNA sequences from known organisms. For example, although the two samples of red algae shown in **Figure 54.13** might appear to be two different species, comparing their sequences of a short standardized section of DNA—a *DNA “barcode”*—to a reference database shows that they belong to the same species. Researchers are increasingly using DNA sequencing for species identification as it becomes cheaper and as DNA sequences from more organisms are added to comparative databases.

Figure 54.13

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    Two samples of one species, the red alga *Chondracanthus
harveyanus.*
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Two species of algae; one with smooth blades and one with blades with jagged edges.

It can also be difficult to census the highly mobile or less visible members of communities, such as microorganisms, deep-sea creatures, and nocturnal species. The small size of microorganisms makes them particularly difficult to sample, so ecologists now commonly use molecular tools to help determine microbial diversity **(Figure 54.14)**.

Figure 54.14

Research Method: Determining Microbial Diversity Using Molecular Tools

Application

Ecologists are increasingly using molecular techniques to determine microbial diversity and richness in environmental samples. One such technique produces a DNA profile for microbial taxa based on sequence variations in the DNA that encodes the small subunit of ribosomal RNA. Researchers used this method to compare the diversity of soil bacteria in 98 habitats across North and South America to help identify environmental variables associated with high bacterial diversity.

Technique

Researchers first extracted and purified DNA from the microbial community in each sample. They used the polymerase chain reaction (PCR; see Figure 20.7) to amplify the ribosomal DNA and labeled it with a fluorescent dye. Restriction enzymes then cut the amplified, labeled DNA into fragments of different lengths, which are separated by gel electrophoresis. (A gel is shown here; see also Figure 20.6.) The number and abundance of these fragments characterize the sample's DNA profile, which was then used to calculate the Shannon diversity index (H) of each sample. Researchers then looked for a correlation between H and several environmental variables, including vegetation type, mean annual temperature and rainfall, and soil acidity.

A researcher examines a DNA sample.

Results

The diversity of bacterial communities was related to soil pH, with the Shannon diversity index being highest in neutral soils (pH 7) and lowest in very acidic soils (pH<5). Amazonian rain forests, which have extremely high plant and animal diversity, had the most acidic soils and the lowest bacterial diversity of the samples tested.

Graph of Shannon diversity versus soil ph. Curve is hump-shaped, with peak Shannon diversity at neutral soil pH levels and lowest diversity in very acidic soils.

Graph of Shannon diversity versus soil ph. The x-axis is soil pH, ranging from 3 to 9 with increments every whole number. The y-axis is Shannon diversity (H) ranging from 2.2 to 3.6 with increments every 0.2. Curve is hump-shaped, with peak Shannon diversity at neutral soil pH levels (pH 7; diversity 3.5) and lowest diversity in very acidic soils (pH 3.5; diversity 2.4). See Table 2 for data summary.

Data from N. Fierer and R. B. Jackson, The diversity and biogeography of soil bacterial communities, *Proceedings of the National Academy of Sciences USA* 103:626–631 (2006).

Diversity and Community Stability

In addition to measuring species diversity, ecologists manipulate diversity in experimental communities in nature and in the laboratory. Many experiments examine the potential benefits of diversity, including increased productivity and stability of biological communities.

Researchers at the Cedar Creek Ecosystem Science Reserve, in Minnesota, have been manipulating plant diversity in experimental communities for more than three decades **(Figure 54.15)**. Higher-diversity communities generally are more productive and are better able to withstand and recover from environmental stresses, such as droughts. More diverse communities are also more stable year to year in their productivity. In one decade-long experiment, for instance, researchers at Cedar Creek created 168 plots, each containing 1,2,4,8, or 16 perennial grassland species. The most diverse plots consistently produced more ****__biomass__ (the total mass of all organisms in a habitat) each year than did the single-species plots.

Figure 54.15

Study plots at the Cedar Creek Ecosystem Science Reserve, site of long-term experiments on manipulating plant diversity.

Field divided into square plots.

Higher-diversity communities are often more resistant to *introduced species*, which are organisms that humans have moved to regions outside of the species' native range. In research conducted off the coast of Connecticut, scientists created communities of different levels of diversity consisting of sessile marine invertebrates, including tunicates (see Figure 34.5). They then examined how vulnerable these experimental communities were to an introduced tunicate. They found that the introduced tunicate was four times more likely to survive in lower-diversity communities than in higher-diversity ones. The researchers concluded that relatively diverse communities captured more of the resources available in the system, leaving fewer resources for the introduced species and thus decreasing its survival.

Trophic Structure

In addition to species diversity, the structure and dynamics of a community also depend on the feeding relationships between organisms—the **trophic structure** of the community. The transfer of chemical energy from its source in plants and other autotrophs (primary producers) through herbivores (primary

consumers) to carnivores (secondary, tertiary, and quaternary consumers) and eventually to decomposers is referred to as a **food chain(Figure 54.16)**. The position an organism occupies in a food chain is called its **trophic level**.

Figure 54.16

Examples of terrestrial and marine food chains.

The figure consists of two parts that illustrate a terrestrial food chain and a marine food chain.

The arrows trace the transfer of the chemical energy in food through the trophic levels of a community when organisms feed on one another. Decomposers, which consume the remains of organisms from all trophic levels, are not shown here.

Part 1: A terrestrial food chain. The steps involved in the food chain are as follows:

1. Primary producer illustrated as a flowering plant.
2. Primary consumer: Herbivore. A grasshopper sitting on the flower.
3. Secondary consumer: Carnivore. A mouse eating the grasshopper.
4. Tertiary consumer: Carnivore. A snake eating the mouse.
5. Quaternary consumer: Carnivore. A flying hawk grabbing the snake.

Part 2: A marine food chain. Steps involved in the food chain are as follows:

1. Primary producers illustrated as different tiny aquatic organisms labeled phytoplankton.
2. Primary consumers illustrated as different zooplankton.
3. Secondary consumers: Carnivore. Two small fishes eating the zooplankton.
4. Tertiary consumers: Carnivore. A large fish eating one small fish.
5. Quaternary consumers: Carnivore. A shark eating one large fish.

VISUAL SKILLS *Suppose the abundance of carnivores that eat zooplankton increased greatly. Use this diagram to infer how that might affect phytoplankton abundance.*

*For suggested answer, see*Appendix A*.*

Food Webs

A food chain is not an isolated unit, separate from other feeding relationships in a community. Instead, a group of food chains are linked together to form a **food web**. Ecologists diagram the trophic relationships of a community using arrows that link species according to who eats whom. In an Antarctic pelagic community, for example, the primary producers are phytoplankton, which serve as food for the dominant grazing zooplankton, especially krill and copepods, both of which are crustaceans **(Figure 54.17)**. These zooplankton species are in turn eaten by various carnivores, including other plankton, penguins, seals, fishes, and baleen whales. Squids, which are carnivores that feed on fishes and zooplankton, are another important link in these food webs, as they are in turn eaten by seals and toothed whales.

Figure 54.17

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An Antarctic marine food web.
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Food web shows multidirectional transfer of food across trophic levels, with phytoplankton at the base of the food web and sperm whales as top-level consumers.

Arrows follow the transfer of the chemical energy in food from the producers (phytoplankton) up through the trophic levels. For simplicity, this diagram omits decomposers. At various times over the last two centuries, humans have also played a role in the Antarctic food web as consumers of fish, krill, and whales.

Food web shows multidirectional transfer of food across trophic levels, described beginning at the level of the primary producer. Phytoplankton is food for copepods and krill. Copepods are food for krill, carnivorous plankton, fish, birds, and squid. Krill are food for carnivorous plankton, birds, fish, crabeater seals, and baleen whales. Carnivorous plankton is food for birds and squid. Birds are food for smaller toothed whales and leopard seals. Fishes are food for birds, squid, elephant seals and leopard seals. Squids are food for elephant seals, fishes, leopard seals, birds, smaller toothed whales, and sperm whales. Crabeater seals are food for smaller toothed whales. Leopard seals are food for smaller toothed whales. Elephant seals are food for smaller toothed whales.

VISUAL SKILLS*For each organism in this food web, indicate the number of other kinds of organisms that it eats. Which two organisms are both predator and prey for each other?*

*For suggested answer, see*Appendix A.**

Figure 54.17 Walkthrough

An Antarctic marine food web

Watch an explanation of this figure in the Figure 54.17 Walkthrough: An Antarctic marine food web.

image

Play

image

How are food chains linked into food webs? A given species may weave into the web at more than one trophic level. In the food web shown in Figure 54.17, krill feed on phytoplankton as well as on other grazing zooplankton, such as copepods. Such “nonexclusive” consumers are also found in terrestrial communities. For instance, foxes are omnivores whose diet includes berries and other plant materials, herbivores such as mice, and other predators, such as weasels. Humans are among the most versatile of omnivores.

Complicated food webs can be simplified in two ways for easier study. First, species with similar trophic relationships in a given community can be grouped into broad functional groups. In Figure 54.17, more than 100 phytoplankton species are grouped as the primary producers in the food web. A second way to simplify a food web is to isolate a portion of the web that interacts very little with the rest of the community. **Figure 54.18** illustrates a partial food web for sea nettles (a type of cnidarian) and juvenile striped bass in the Chesapeake Bay estuary on the Atlantic coast of the United States.

Figure 54.18

Partial food web for Chesapeake Bay estuary.

Partial food web includes fish eggs, sea nettle, zooplankton, fish larvae and juvenile striped bass.

The sea nettle (Chrysaora quinquecirrha) and the juvenile striped bass (Morone saxatilis) are the main predators of fish larvae.

Partial food web includes fish eggs, sea nettle, zooplankton, fish larvae and juvenile striped bass. Fish eggs are food for sea nettle. Zooplankton are food for sea nettle and fish larvae. Fish larvae (interactions highlighted) are food for sea nettle and juvenile striped bass.

VISUAL SKILLS *Based on this food web, identify organisms that function as primary, secondary, or tertiary consumers.*

*For suggested answer, see*Appendix A.**

HHMI Biointeractive Video

Riverine Food Webs: How Flow Rates Affect Biomass

image

image

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Limits on Food Chain Length

Each food chain within a food web is usually only a few links long. In the Antarctic web of Figure 54.17, there are rarely more than seven links from the producers to any top-level predator, and most chains in this web have fewer links. In fact, most food webs studied to date have chains consisting of five or fewer links.

Why do food chains tend to be relatively short? The most common explanation, known as the **energetic hypothesis**, suggests that the length of a food chain is limited by the inefficiency of energy transfer along the chain. On average, only about 10% of the energy stored in the organic matter of each trophic level is converted to organic matter at the next trophic level (see Concept 55.3). Thus, a producer level consisting of 100kg of plant material can support about 10kg of herbivore biomass and 1kg of carnivore biomass. The energetic hypothesis predicts that food chains should be relatively longer in habitats characterized by higher photosynthetic production, since the amount of energy stored in primary producers is greater than in habitats with lower photosynthetic production.

Ecologists tested the energetic hypothesis in experiments that mimicked the “tree-hole” communities found in tropical forests. Many trees have small branch scars that rot, forming holes in the tree trunk. These tree holes hold water and provide a habitat for tiny communities consisting of microorganisms and insects that feed on leaf litter, as well as predatory insects. **Figure 54.19** shows the results of experiments in which researchers manipulated productivity by varying the amount of leaf litter in an experiment using artificial tree holes (water-filled pots placed around the trees); previous studies had shown that the communities that colonized these pots were similar to those in natural tree holes. As predicted by the energetic hypothesis, the pots with the most leaf litter, and hence the greatest total food supply at the producer level, supported the longest food chains.

Figure 54.19

Test of the energetic hypothesis for the restriction of food chain length.

Bar chart for the number of trophic links, ranging from 0 to 5, versus high, medium and low productivity.

Researchers manipulated the productivity of experimental tree-hole communities in Queensland, Australia, by providing leaf litter input at three levels. Reducing energy input reduced food chain length, a result consistent with the energetic hypothesis.

Bar chart for the number of trophic links, ranging from 0 to 5, versus high, medium and low productivity. With high productivity (control; the natural rate of litter fall), 5 trophic links are shown. With medium

productivity (one-tenth the natural rate), 3 trophic links are shown. With low productivity (one-one hundredth the natural rate), 1 trophic link is shown.

Another factor that may limit food chain length is that carnivores tend to be larger at higher trophic levels. The size of a carnivore puts an upper limit on the size of food it can take into its mouth. And except in a few cases, large carnivores cannot live on very small food items because they cannot obtain enough food in a given time to meet their metabolic needs. Among the exceptions are baleen whales, huge filter feeders with adaptations that enable them to consume enormous quantities of krill and other small organisms (see Figure 41.5).

Species with a Large Impact

Certain species have an especially large impact on the structure of entire communities because they are highly abundant or play a pivotal role in community dynamics. The impact of these species occurs through trophic interactions and their influence on the physical environment.

Foundation species have strong effects on their communities as a result of their large size or high abundance. Examples of foundation species include trees, desert shrubs such as creosote, and certain marine algae, such as kelp. Such species often have community-wide effects because they provide habitat or food. Foundation species may also be competitively dominant—superior in exploiting key resources such as space, water, nutrients, or light.

The impact of a foundation species can be discovered when it is removed from the community. For example, the American chestnut was a competitively dominant tree in deciduous forests of eastern North America before 1910, making up more than 40% of mature trees. Then humans accidentally introduced the fungal disease chestnut blight to New York City via nursery stock imported from Asia. Between 1910 and 1950, this fungus killed almost all of the chestnut trees in eastern North America. In this case, removing the foundation species had a relatively small impact on some species but large effects on others. Oaks, hickories, beeches, and red maples that were already present in the forest increased in abundance and replaced the chestnuts. No mammals or birds seemed to have been harmed by the loss of the chestnut, but seven species of moths and butterflies that fed on the tree became extinct.

In contrast to foundation species, **keystone species** are not usually abundant in a community. They exert strong control on community structure not by numerical might but by their pivotal ecological roles. **Figure 54.20** highlights the importance of a keystone species, a sea star, in maintaining the diversity of an intertidal community. In this case, the sea star affects its community by feeding on and there by limiting the abundance of a competitively dominant species, a mussel.

Figure 54.20

Inquiry: Is *Pisaster ochraceus* a keystone species?

Experiment

In rocky intertidal communities of western North America, the relatively uncommon sea star *Pisaster ochraceus* preys on mussels such as *Mytilus californianus*, a dominant competitor for space. Robert Paine, of the University of Washington, removed *Pisaster* from an area in the **intertidal zone** and examined the effect on species richness.

Pisaster feeds on a mussel.

Results

In the absence of *Pisaster*, species richness declined as mussels monopolized the rock face and eliminated most other invertebrates and algae. In a control area where *Pisaster* was not removed, species richness changed very little.

Graph of number of species present versus year, with and without Pisaster. Without Pisaster present, the number of species present declines sharply, from about 17 to about 2.

Graph of number of species present versus year, with and without Pisaster. The x-axis is year, ranging from 1963 to 1973 with increments every whole year. The y-axis is number of species present, ranging from 0 to 20 with increments every 5 species. With Pisaster present (control), the number of species present remains relatively constant across all years, from about 17 to 20. Without Pisaster present (experimental), the number of species present plunges from 17 to about 2 between 1963 and 1966, and remains at about 2 for the duration of the experiment (to 1973).

Conclusion

Pisaster acts as a keystone species, exerting an influence on the community that is not reflected in its abundance.

Data from R. T. Paine, Food web complexity and species diversity, *American Naturalist* 100:65–75 (1966).

WHAT IF?***Suppose that an invasive fungus killed most individuals of* Mytilus *at these sites. Predict how species richness would be affected if* Pisaster *were then removed.*

*For suggested answer, see*Appendix A.**

Still other organisms exert their influence on a community not through trophic interactions but by changing the physical environment. Species that create or dramatically alter their environment are called **ecosystem engineers**. A familiar ecosystem engineer is the beaver **(Figure 54.21)**. The effects of ecosystem engineers on other species can be positive or negative, depending on the needs of the other species. Some foundation species, such as trees, can also be considered ecosystem engineers because their presence modifies the physical environment in ways that create habitats on which other species depend.

Figure 54.21

Beavers as ecosystem engineers.

Two beavers build a dam.

By felling trees, building dams, and creating ponds, beavers can transform large areas of forest into flooded wetlands.

Bottom-Up and Top-Down Controls

The ways in which adjacent trophic levels affect one another can be useful for describing community organization. These effects occur in two general ways: organisms can be controlled by what they eat (“bottom-up” control) or by what eats them (“top-down” control).

In **bottom-up control**, the abundance of organisms at each trophic level is limited by nutrient supply or the availability of food at lower trophic levels. As shown on the left side of **Figure 54.22**, the supply of nutrients controls plant numbers, which control herbivore numbers, which in turn control predator numbers. To change the structure of a bottom-up community, you would need to alter the biomass or abundance of organisms at lower trophic levels, allowing those changes to propagate up through the food web. For example, if you add nutrients to stimulate plant growth, then each of the higher trophic levels should also increase in biomass. If you change predator abundance, however, the effect should not extend down to the lower trophic levels.

Figure 54.22

Bottom-up and top-down control.

Two terrestrial and marine food chains illustrate bottom-up and top-down control.

The abundance of organisms at each trophic level can be controlled from the bottom up (by the availability of food or nutrients) or from the top down (by rates of consumption at higher trophic levels). Large gray arrows indicate the direction of control in each of the food chains shown.

The terrestrial chain illustrates bottom-up control, from primary producer through tertiary consumers. Small black arrows represent the transfer of food up through the trophic levels of a food chain. A limiting nutrient supply or the availability of food at lower trophic levels limits the abundance of organisms at higher trophic levels. The marine chain illustrates top-down control in the opposite direction. The abundance of organisms at higher trophic levels controls the abundance of organisms at lower trophic levels.

In **top-down control**, the abundance of organisms at each trophic level is controlled by the abundance of consumers at higher trophic levels. Thus, in top-down control, predators limit herbivores, and herbivores limit plants or algae. In the example on the right side of Figure 54.22, removing the top carnivores (killer whales) should increase the abundance of primary carnivores (sea otters), decrease the number of herbivores (urchins), and increase the abundance of primary producers (kelp). The effects thus move down the trophic structure as alternating +/– effects.

Ecologists have applied top-down control to improve water quality in lakes with a high abundance of algae. For example, in lakes with three trophic levels, removing fish should improve water quality by increasing zooplankton density, thereby decreasing algal populations **(Figure 54.23)**. In lakes with four trophic levels, adding top predators should have the same effect.

Figure 54.23

Restoring water quality in a lake with top-down control.

Two diagrams of lake condition before and after restoration. Before restoration, fish were abundant, zooplankton was rare, and algae were abundant. After restoration, fish are rare, zooplankton is abundant, and algae is rare.

Decreasing the abundance of fish that ate zooplankton results in a decrease in the biomass of algae, improving water quality. Arrow thickness indicates the relative strength of each top-down control.

Ecologists in Finland used this approach to help purify Lake Vesijärvi, a large lake that was polluted with city sewage and industrial waste until 1976. After pollution controls reduced these inputs, the lake's water quality began to improve. In 1986, however, massive blooms of cyanobacteria began occurring. These blooms coincided with an increase in the population of roach, a fish species that eats zooplankton, which otherwise keep cyanobacteria in check. To reverse these changes, ecologists removed nearly a million kilograms of fish from the lake between 1989 and 1993, reducing roach abundance by about 80%. At the same time, they added a fourth trophic level by stocking the lake with pike perch, a predatory fish that eats roach. The water cleared up, and the last cyanobacterial bloom was in 1989. Ecologists continue to monitor the lake for cyanobacterial blooms and low oxygen availability, but the lake has remained clear, even though roach removal ended in 1993.

HHMI BioInteractive Video

Some Animals Are More Equal Than Others: Keystone Species and Trophic Cascades

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Trophic Cascades in Salt Marsh Ecosystems

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